S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

SENTINEL

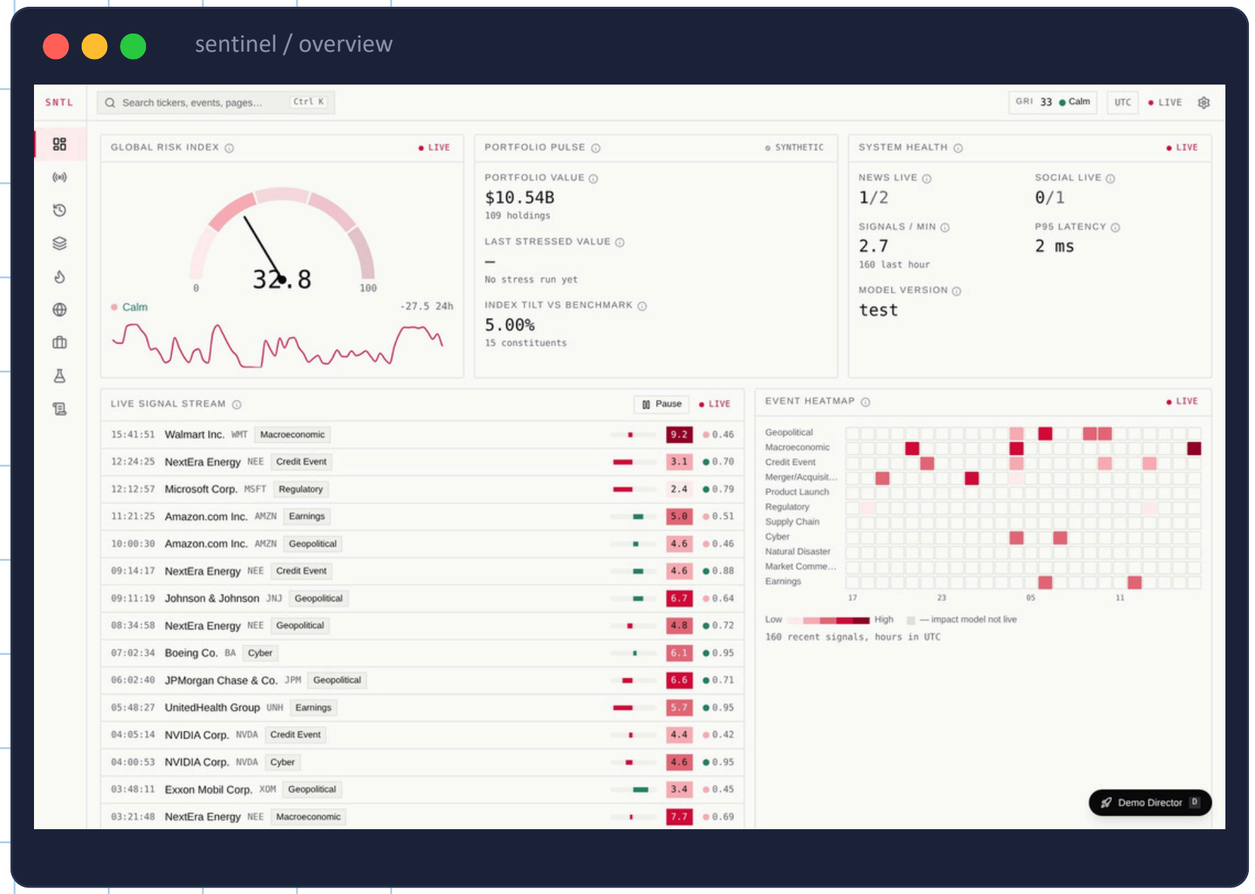
Risk intelligence with a memory.

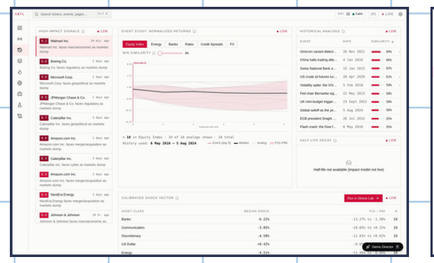
Turns breaking financial headlines into quantified, auditable portfolio actions, by remembering what happened the last time.

139	10	100%
automated tests pass	live terminal views	public + synthetic data

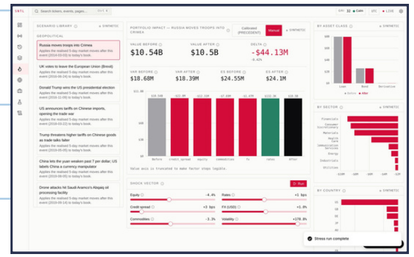
Indira Gandhi Delhi Technical University for Women (IGDTUW)

chaitenya013mca25@igdtuw.ac.in · Individual submission

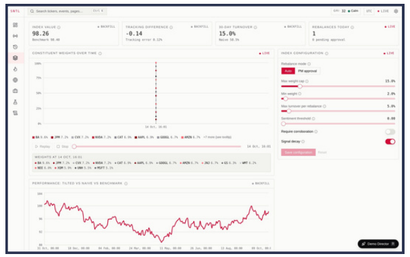




PRECEDENT analogs



Stress Lab



Adaptive Index

Real captures from the running app on seeded demo data

Headlines move portfolios, but analysts get noise without memory

THE GAP TODAY

Noise overwhelms signal

Only 29 of 150 hand-labelled headlines (19%) were risk-relevant. Analysts triage five items for every one that matters.

Alerts have no memory

Sentiment tools say how today's headline feels, not what markets did the last time something similar hit.

Insight stops at the dashboard

Nothing links an event to portfolio loss, to a rule that fires, and to a decision someone can audit.

ONE LOOP, FOUR VERBS

Detect

Relevance gate, event type, FinBERT sentiment and an integrity score turn raw headlines into scored signals.

NLP pipeline

Remember

PRECEDENT finds point-in-time analog events and shows what markets did over the next 20 days.

PRECEDENT

Quantify

Calibrated shocks revalue the whole book: VaR/ES, Shapley attribution, reverse stress.

Stress Lab

Act

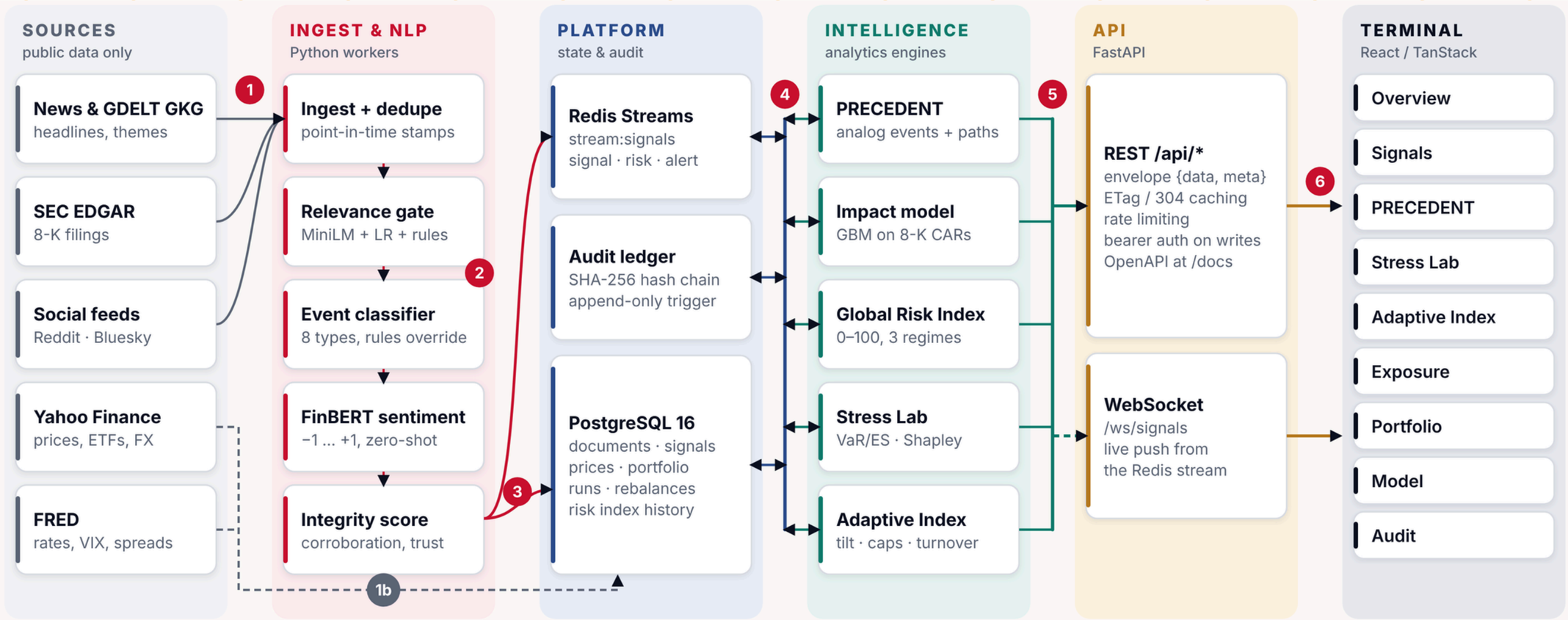
Adaptive Index tilts weights within caps; a PM approves; every step is hash-chained.

Index + Audit

Typical workflow	SENTINEL
Keyword alerts	**Learned + rule hybrid gate (AUC 0.78 vs 0.56 rules-only)**
Today's sentiment only	**Sentiment plus analog outcomes, with a low-n warning**
Spreadsheet what-ifs	**Full revaluation of 109 holdings in about 0.2 s, with attribution**
Rebalance by hand (58.5% turnover*)	**Constrained tilt, optional PM approval (15.0% turnover*)**

**30-day backfill on synthetic demo data. AUC from the held-out set of 150 labelled headlines.*

One event stream feeds five engines in one auditable platform



CROSS-CUTTING

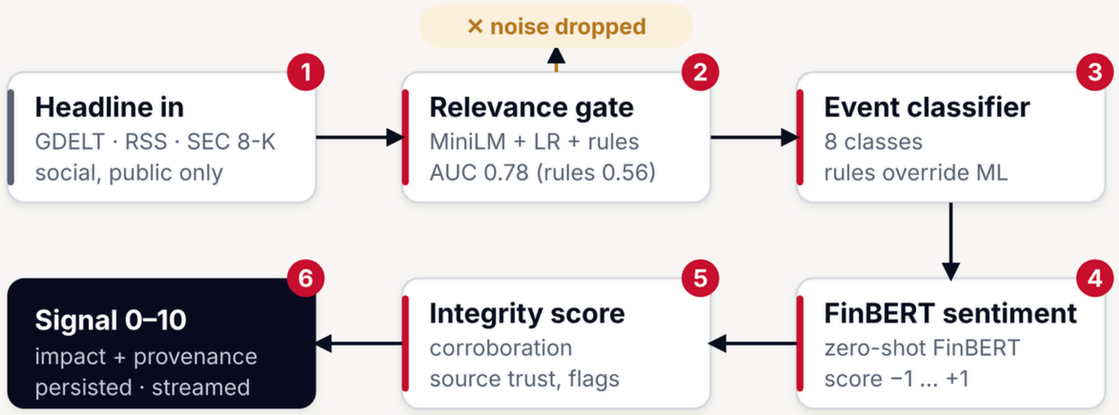
Provenance tag on every row: live · replay · backfill · synthetic · Point-in-time analogs, no look-ahead · APScheduler jobs · Docker Compose · 139 automated tests

1. Public sources ingested, deduped, time-stamped
2. NLP chain scores relevance, type, sentiment, integrity
3. Signal saved to Postgres, published on Redis Streams
4. Engines read and write state via the platform
5. REST and WebSocket serve the terminal
6. Terminal renders 9 views from real API data

Four engines, each built to be inspected, tested and explained

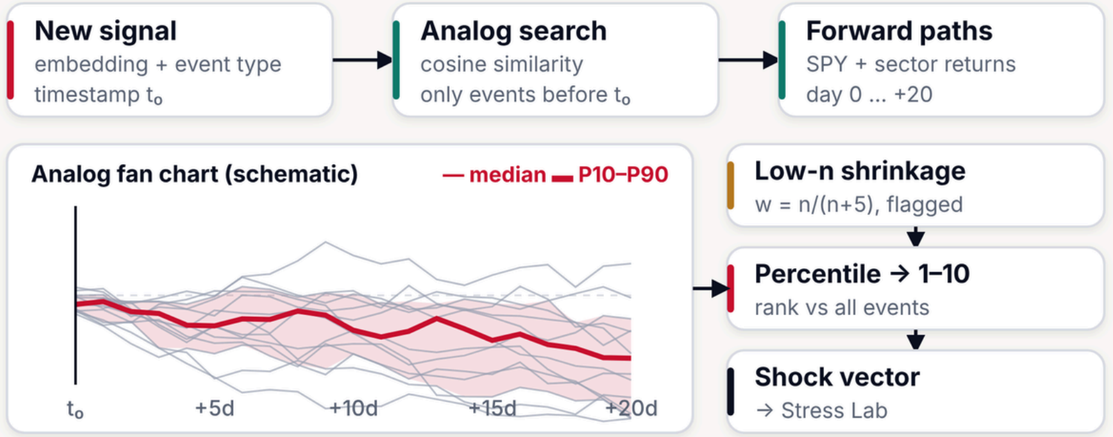
1 NLP signal pipeline

MiniLM + FinBERT · 8 event types · hybrid gate



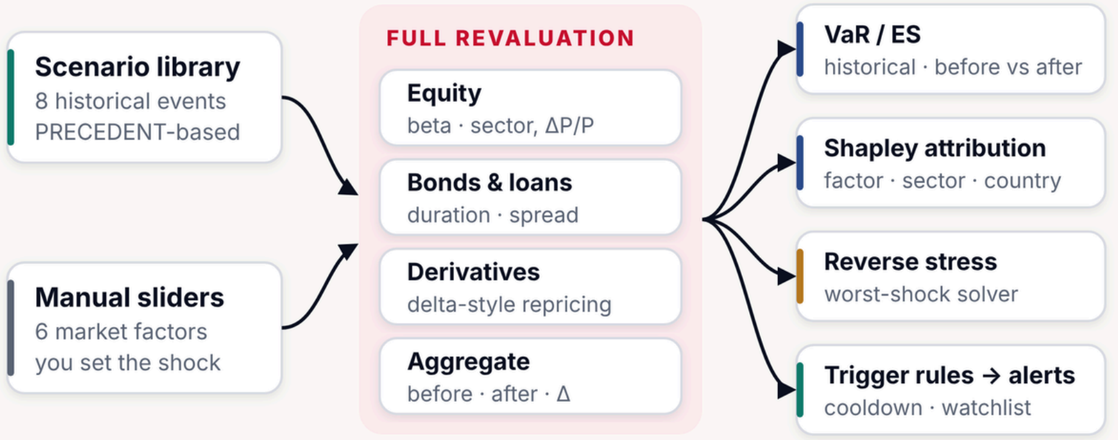
2 PRECEDENT analog engine

point-in-time · shrinkage w = n/(n+5)



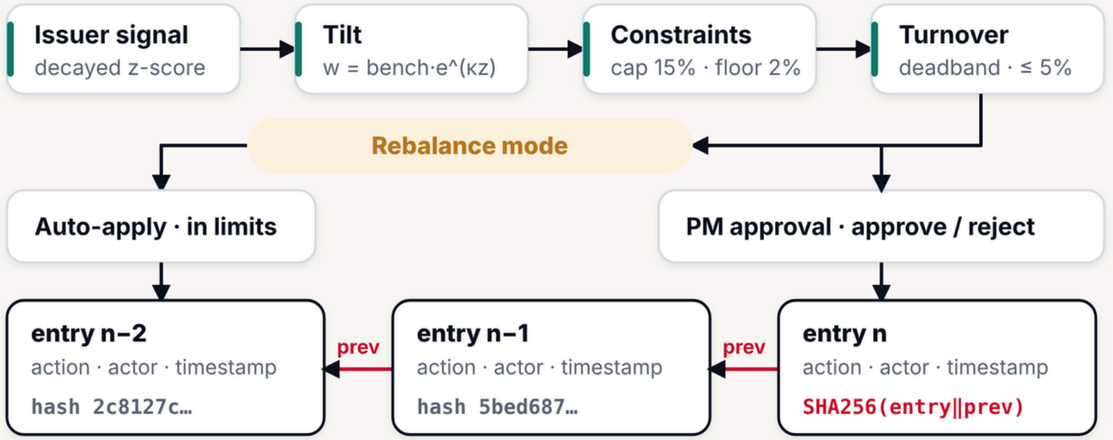
3 Stress Lab

full revaluation · Shapley · reverse stress

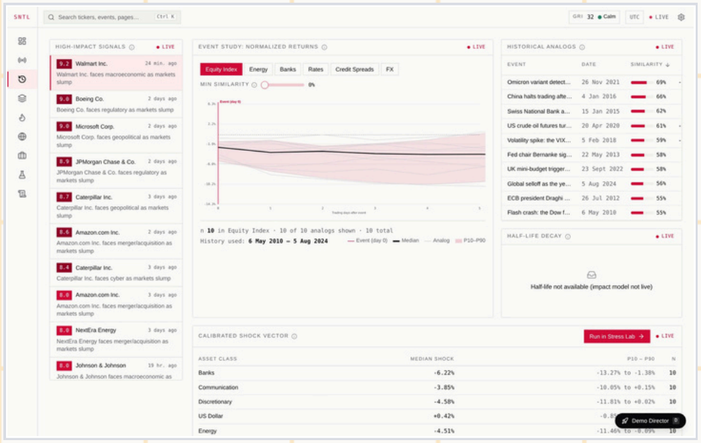


4 Adaptive Index + audit ledger

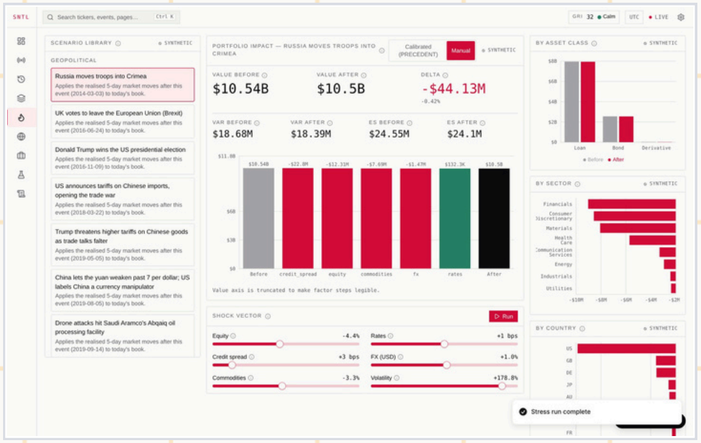
caps · deadband · SHA-256 hash chain



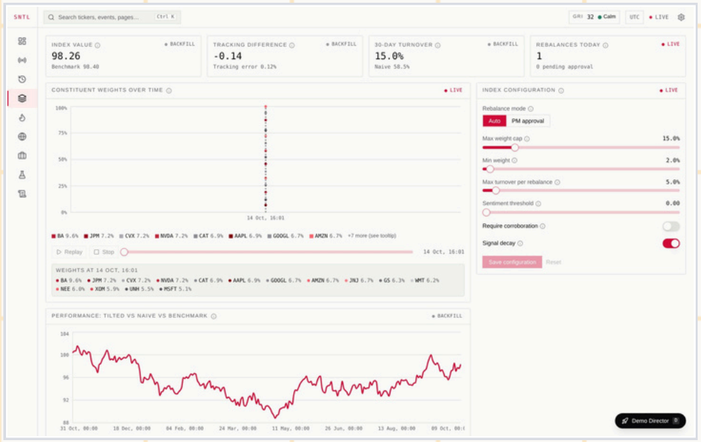
Measured, not claimed: beats naive baselines, with honest error bars



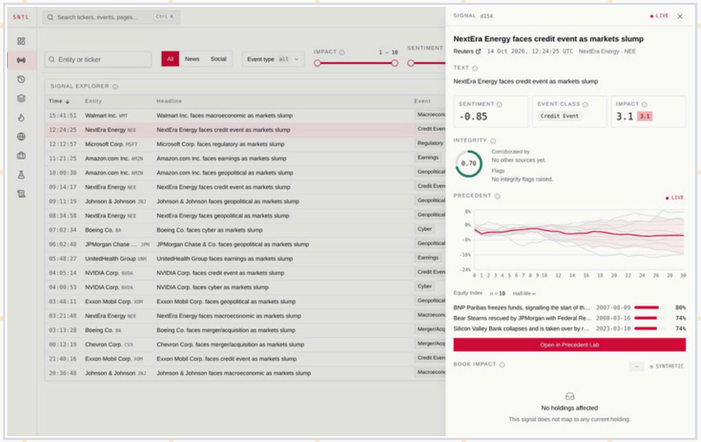
PRECEDENT: analog events, fan chart, calibrated shocks



Stress Lab: Crimea scenario, about −$44M on a $10.5B book



Adaptive Index: weights, 15.0% turnover vs 58.5% naive



Signals: drawer with integrity, analogs, book impact

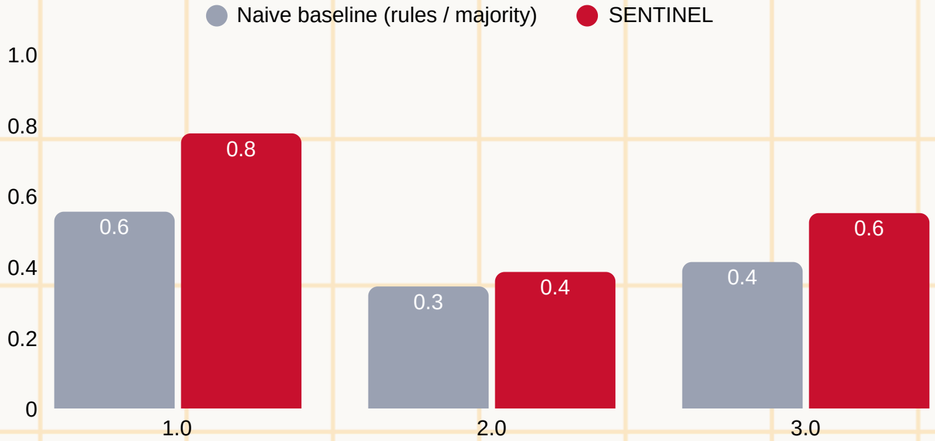
0.78
Relevance AUC
naive rules-only: 0.56

55%
Sentiment accuracy
majority-class baseline: 41%

15%
Index turnover
naive rebalance: 58.5%*

0.2 s
Full stress run
109 holdings, VaR/ES + Shapley

SENTINEL vs naive baseline (held-out, n = 150)



Screenshots are the running app on seeded synthetic demo data (portfolio, prices, signals). The build sandbox had no live-feed access; replay mode uses archived GDELT episodes.

*Held-out set: 29 relevant of 150, F1 95% CI 0.22–0.54. Impact model (5,825 8-K events): MAE 0.0187 vs 0.0195 baseline, Spearman 0.33. *30-day backfill, synthetic prices.*

From headline to defensible decision, for ratings, index and risk teams

Credit & ratings surveillance analyst

Today Scans feeds and judges severity from memory.
With SENTINEL Relevance-gated signals ranked 1–10, with historical analogs and an integrity score.

Earlier, evidence-backed watchlist calls

Index and portfolio manager

Today Rebalances by hand after events.
With SENTINEL Signal-tilted weights inside caps, optional PM approval. Demo: 15% vs 58.5% naive turnover.

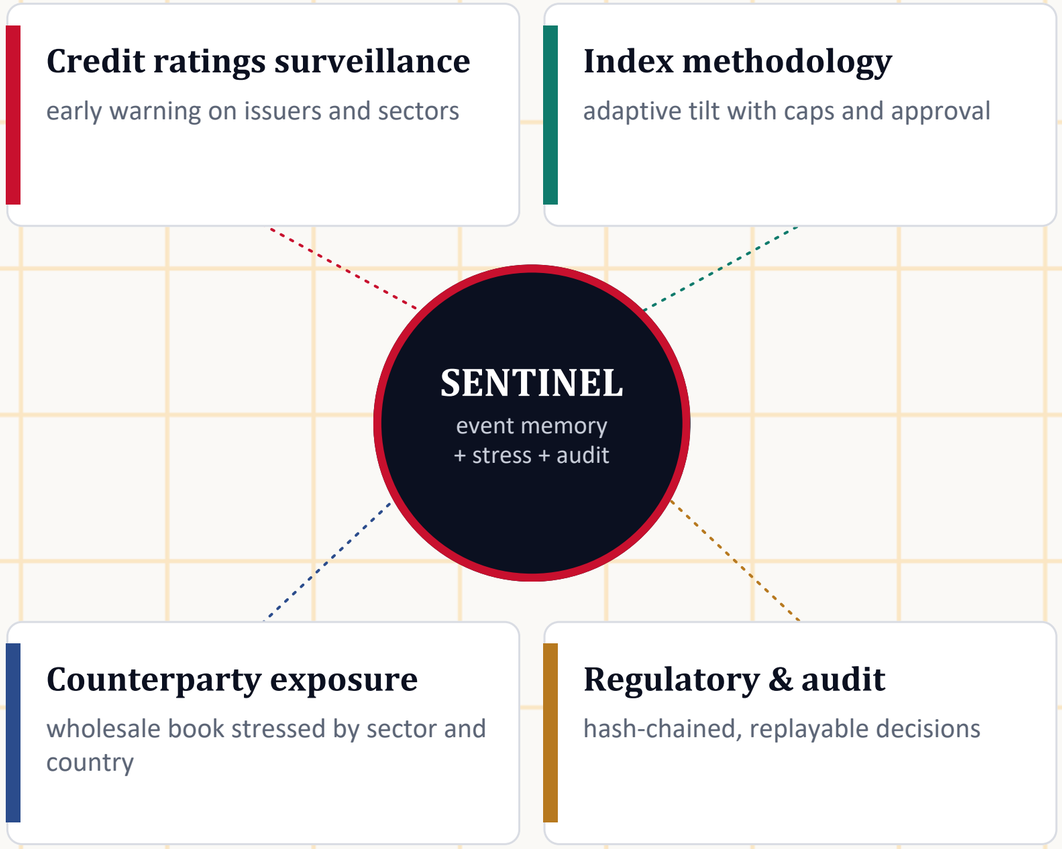
Lower turnover, same discipline

Risk & compliance officer

Today Runs what-ifs in spreadsheets; hard to prove who changed what.
With SENTINEL Stress runs with VaR/ES and Shapley attribution; audit chain verifies 362 entries in about 17 ms.

Audit-ready by design

WHERE IT PLUGS IN



Illustrative mapping of capabilities to workflows; not a statement about any company's internal systems.

Honest limits today, a clear path to production tomorrow

WHAT IS NOT SOLVED YET

Small gold-label set
150 held-out headlines (29 relevant): relevance F1 95% CI 0.22–0.54. Directional, not definitive.

Learned event classifier trails rules
Macro-F1 0.32 vs 0.47 held-out, so rule overrides stay in the decision.

Modest impact-model lift
MAE 0.0187 vs 0.0195 baseline, Spearman 0.33; 8-K events skew to “Other”.

Synthetic book, illustrative migration matrix
The wholesale portfolio is synthetic and rating migration is a documented assumption.

Live connectors not exercised in the build
Yahoo, FRED, SEC and GDELT code paths exist; the demo runs on replay and synthetic data.

NEXT 90 DAYS

Month 1
Label 1,000+ headlines; recalibrate relevance and event models.

Month 2
Verify live feeds; run the PSI drift monitor with alerts.

Month 3
Real rating-transition data; SciPy SLSQP worst-shock solver.

Beyond
Grounded LLM analyst narrative with a citation guard; multi-tenant auth.

Responsible use Outputs are risk analytics for demonstration, not investment advice. Built with AI assistance (Claude); data choices, validation and claims are the author's. Public and synthetic data only.